

ECE253/CSE208 Introduction to Information Theory

Lecture 6: Asymptotic Equipartition Property (AEP)

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- Chap 2 of *Elements of Information Theory (2nd Edition)* by Cover–Thomas
- Lecture 1 of *Information Theory for Wireless Comms.* by Saif Mohammed
- Chap 2 of *Network Information Theory* by Gamal–Kim.

Law of Large Numbers (LLN) in Information Theory

AEP

Weak Typicality

AEP Properties

Strong
Typicality

Empirical
Entropy

Joint Typicality

- LLN in probability (Sample mean converges to the true mean):

$$\bar{X} = \frac{1}{n} \sum_i X_i \xrightarrow[n \rightarrow \infty]{\text{i.p.}} \mathbb{E}(X).$$

- LLN in info theory (Normalized self-information (per symbol) converges to the true entropy)

$$\bar{H}(X) = \frac{1}{n} \log \frac{1}{p(X_1, \dots, X_n)} \xrightarrow[n \rightarrow \infty]{\text{i.p.}} H(X).$$

- For notation simplicity, let $X_1^n = (X_1, X_2, \dots, X_n)$ and $x_1^n = (x_1, x_2, \dots, x_n)$ denote the sequence of those n random variables and outcomes, respectively.

Asymptotic Equipartition Property (AEP)

AEP

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Theorem (AEP)

If $\{X_i\}_{i=1}^n \sim p(x)$ are i.i.d., then $-\frac{1}{n} \log p(X_1^n) \xrightarrow{i.p.} H(X)$.

Proof: by the weak law of large numbers (WLLN), we have

$$-\frac{1}{n} \log p(X_1^n) = -\frac{1}{n} \sum_i \log p(X_i) \xrightarrow{i.p.} -\mathbb{E}[\log p(X)] = H(X)$$

Remark:

- Shannon–McMillan–Breiman Theorem: For stationary ergodic sources (much more general than i.i.d.) AEP holds almost surely.
- We restrict to i.i.d. sources here since convergence in probability is sufficient for the achievability part of the Source Coding Theorem.

An Example of AEP

AEP

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Example (Sanity check of AEP)

Consider i.i.d. $\{X_i\}_{i=1}^n \sim \text{Bern}(p)$, let $q = 1 - p$. We have

$$p(x_1^n) = p^{\sum_{i=1}^n x_i} \times q^{n - \sum_{i=1}^n x_i} \xrightarrow{i.p.} p^{np} q^{nq}.$$

$$H(X) = -p \log p - q \log q \implies -nH(X) = \log(p^{np} q^{nq})$$

This matches the AEP: $p(X_1^n) \xrightarrow{i.p.} 2^{-nH(X)}$.

Q: AEP is based on the assumption that X_1^n are i.i.d. How about for non-iid case?

A: Entropy rate of stochastic processes; see the next lecture (Chap 4).

Typical Sequences

AEP

Weak Typicality

AEP Properties

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Example

Consider i.i.d. $\{X_i\}_{i=1}^n \sim \text{Bern}(p)$, then $p(x_1^n) = \prod_{i=1}^n p(x_i)$. For example, $p(1, 0, 1, 1, 0, 1) = p^4(1-p)^2$. Clearly, not all sequences are generated equally.

We see that $p(X_1^n)$ is close to $2^{-nH(X)}$ with high probability. That is, the probability $p(X_1^n)$ assigned to an observed sequence is close to $2^{-nH(X)}$.

Almost all sequences have probability $\approx 2^{-nH(X)}$.

$$\Pr\left\{X_1^n : p(X_1^n) = 2^{-n(H \pm \epsilon)}\right\} \approx 1,$$

if X_1^n are i.i.d. $\sim p(x)$.

Typical Set

AEP

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We can thus divide the set of all sequences into two classes:

1. **Typical set**, where the probability of each typical sequence is close to $2^{-nH(X)}$.
2. **Atypical set** that contains all the other sequences.
3. Typical set is primarily a theoretical tool that is defined to help prove some theorems, even though its concept is somehow counter-intuitive, as we will see later.

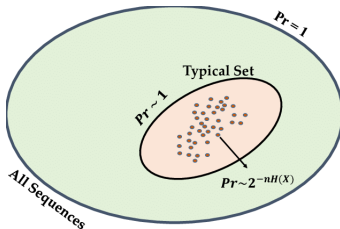


Figure: Typical sequences and typical set.

Weakly Typical Sequences

AEP

Weak Typicality

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Joint Typicality

Some sequences are “typical” in the sense that their information is about the same as the self-information expected. We define those **weakly typical sequences** as follows:

Definition (ϵ -typical sequence and ϵ -typical set)

A sequence $x_1^n \in \mathcal{X}^n$ is an ϵ -typical sequence with respect to $p(x)$ if

$$2^{-n(H(X)+\epsilon)} \leq p(x_1^n) \leq 2^{-n(H(X)-\epsilon)}.$$

Further, a typical set $A_\epsilon^{(n)}$ is the set containing all ϵ -typical sequences $x_1^n \in \mathcal{X}^n$.

Remark: Weak typicality applies to both discrete and continuous alphabets. Strong typicality requires finite alphabets.

Interpretation: Typical sequences are those whose empirical distribution is close to the true distribution in KL divergence.

Implications of Typical Sequences

AEP

Weak Typicality

AEP Properties

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Joint Typicality

- Intuitively, we would like to assign shorter bit strings to the 'typical' sequences to reduce the expected length of the code. It turns out that as n grows large, almost all the probability concentrates on the typical set.
- Typical set is made by all the sequences that are giving us an amount of information close to the average information of the source distribution.
- The least and most probable sequences give us less information than the average.
- Hence, AEP filters out a lot of highly unlikely sequences as well as a small number of highly likely sequences.

Examples of Typical Sequences (1/3)

AEP

Weak Typicality

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The “typicality” is in the sense of *normalized self-information close to the true entropy*, rather than “most likely”.

Example (Most likely sequence is often not in the typical set)

- For i.i.d. $X_i \sim \text{Bern}(0.9)$, $H(X) = 0.469$. The most likely sequence of outcome is the sequence of all 1's:

$$-\frac{1}{n} \log_2 p(x_1^n = (1, 1, \dots, 1)) = -\frac{1}{n} \log_2(0.9^n) = 0.152.$$

Hence, for small enough ϵ , all-one sequence is not in the typical set.

Examples of Typical Sequences (2/3)

AEP

Weak Typicality

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For Bernoulli random variables, the typical set consists of sequences with average numbers of 0's and 1's in n independent trials. Because if a sequence has np 1's and nq 0's for n trials, then $p(x_1^n) = p^{np}q^{nq} \implies$

$$-\frac{1}{n} \log_2 p(x_1^n) = -p \log p - q \log q = H(X).$$

- If $p = 0.9, n = 10$, then the typical set consist of all sequences that have a single 0 in the entire sequence.
- If $p = 0.5$, then every possible binary sequences belong to the typical set.

Examples of Typical Sequences (3/3)

AEP

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Example

A computer program generates a binary sequence of length 12 digits (i.i.d. $X_i \sim \text{Bern}(\frac{1}{3})$). One of the following four sequences is generated from the program. Which one is it?

(a) 0 0 0 0 0 0 0 0 0 0 0 0, $\Pr(a) = (2/3)^{12} = 7.7 \times 10^{-3}$

(b) 1 0 1 1 0 1 0 1 0 1 0 0, $\Pr(b) = (2/3)^6 \times (1/3)^6 = 1.2 \times 10^{-4}$

(c) 0 0 0 1 0 0 0 1 0 0 1 0, $\Pr(c) = (2/3)^9 \times (1/3)^3 = 9.6 \times 10^{-4}$

(d) 1 1 1 1 1 1 1 1 1 1 1 1, $\Pr(d) = (1/3)^{12} = 1.9 \times 10^{-6}$.

The answer is sequence (c), although sequence (a) has a higher probability of occurrence. An intuition based reasoning is that, since the source outputs are i.i.d., roughly $1/3$ of the 12 digits should be one and $2/3$ should be zero. This is in fact true as the length of the sequence is increasing. Those sequences are called “typical sequences”.

Counting the Typical Ones

AEP

Weak Typicality

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- Consider a random source i.i.d. $X_i \sim \text{Bern}(p)$ generating a sequence of length n .
- There are $\binom{n}{np}$ independent sequences that have exactly np ones and the probability of each such sequence is $p^{np}(1-p)^{n(1-p)}$.
- Approximate $\binom{n}{np}$ by using the Stirling's formula: $n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$.

$$\begin{aligned}\log \binom{n}{np} &= \log \left(\frac{n!}{(np)!(n-np)!} \right) \\ &\approx -\log \left(\sqrt{2\pi np(1-p)} \right) - n \times [p \log p + (1-p) \log(1-p)]\end{aligned}$$

$$\Rightarrow \boxed{\binom{n}{np} \approx \frac{2^{nH(p)}}{\sqrt{2\pi np(1-p)}}}$$

Hence, the number of such sequences increases as $2^{nH(p)}$, but it's a much smaller subset of all possible sequences.

Properties of AEP

AEP

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Theorem

We have the following properties of the AEP:

1. If $x_1^n \in A_\epsilon^{(n)}$ then $H(X) - \epsilon \leq -\frac{1}{n} \log p(x_1^n) \leq H(X) + \epsilon$.
2. $\Pr \left\{ A_\epsilon^{(n)} \right\} > 1 - \epsilon$ *for n sufficiently large*.
3. $\left| A_\epsilon^{(n)} \right| \leq 2^{n(H(X) + \epsilon)}$.
4. $\left| A_\epsilon^{(n)} \right| \geq (1 - \epsilon) 2^{n(H(X) - \epsilon)}$ *for n sufficiently large*.

The above Theorem asserts that i) the typical set has probability nearly 1; ii) all elements in it are nearly equiprobable; and iii) the size of the typical set is about $2^{nH(X)}$.

Implications of AEP Properties (1/2)

AEP

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For small ϵ , we have $|A_\epsilon^{(n)}| \approx 2^{nH(X)}$. Thus, the fraction of sequences that are typical is

$$\rho_n := \frac{|A_\epsilon^{(n)}|}{|\mathcal{X}^n|} \approx \frac{2^{nH(X)}}{|\mathcal{X}|^n} = \frac{2^{nH(X)}}{2^{n \log |\mathcal{X}|}} = 2^{-n(\log |\mathcal{X}| - H(X))}.$$

- For non-uniform distribution: $H(X) < \log |\mathcal{X}|$, $\rho_n \rightarrow 0$ as $n \rightarrow \infty$.
- For uniform distribution: $H(X) = \log |\mathcal{X}| \rightarrow \rho_n = 1$, every sequence is typical.

Implications of AEP Properties (2/2)

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Everything outside the typical set has a negligible probability.

- The typical set $A_\epsilon^{(n)}$ contains an exponentially small fraction of all sequences, yet it carries almost all the probability: $\Pr\{A_\epsilon^{(n)}\} > 1 - \epsilon$.
- Thus, a randomly generated sequence lies in the typical set with high probability, despite the set being much smaller than the space of all sequences.
- For sufficiently large n , X_1^n behaves as if it were drawn nearly uniformly from $A_\epsilon^{(n)}$, a manifestation of the [asymptotic equipartition property](#).

Proof of AEP Properties (1/3)

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Proof.

1. From the definition of ϵ -typical sequences, if $x_1^n \in A_\epsilon^{(n)}$, then

$$2^{-n(H(X)+\epsilon)} \leq p(x_1^n) \leq 2^{-n(H(X)-\epsilon)}.$$

Taking the logarithm and dividing by $-n$ yields Property 1.



Proof of AEP Properties (2/3)

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Proof.

1. Since

$$-\frac{1}{n} \log p(X_1^n) \xrightarrow{\text{i.p.}} H(X),$$

for any $\delta > 0$, there exists n_0 such that for all $n \geq n_0$,

$$\Pr\left(\left|-\frac{1}{n} \log p(X_1^n) - H(X)\right| < \epsilon\right) > 1 - \delta.$$

The event

$$\left|-\frac{1}{n} \log p(X_1^n) - H(X)\right| < \epsilon$$

holds if and only if X_1^n is ϵ -typical. Setting $\delta \triangleq \epsilon$ yields Property 2.



Proof of AEP Properties (3/3)

AEP

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Proof.

3. By Property 1, we have $2^{-n(H(X)+\epsilon)} \leq p(x_1^n)$, $\forall x_1^n \in A_\epsilon^{(n)} \implies$

$$2^{-n(H(X)+\epsilon)} |A_\epsilon^{(n)}| \leq \sum_{x_1^n \in A_\epsilon^{(n)}} p(x_1^n) \leq \sum_{x_1^n \in \mathcal{X}^n} p(x_1^n) = 1,$$

which proves Property 3.

4. By Property 2, for n sufficiently large, we have $\Pr \{A_\epsilon^{(n)}\} > 1 - \epsilon$. Hence,

$$1 - \epsilon < \Pr \{A_\epsilon^{(n)}\} = \sum_{x_1^n \in A_\epsilon^{(n)}} p(x_1^n) \leq \sum_{x_1^n \in A_\epsilon^{(n)}} 2^{-n(H(X)-\epsilon)} = 2^{-n(H(X)-\epsilon)} |A_\epsilon^{(n)}|.$$

The “ $\leq$ ” follows from the upper bound of $p(x_1^n)$.

Strong Typicality

AEP

Weak Typicality

AEP Properties

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Joint Typicality

Definition (Strongly Typical Sets)

Let $X_1, \dots, X_n \sim p(x)$ be i.i.d. random variables with finite alphabet $\mathcal{X}$ and probability mass function p . For $\delta > 0$, the strongly typical set $T_\delta^{(n)}$ is defined as

$$T_\delta^{(n)} = \left\{ x_1^n \in \mathcal{X}^n : \sum_{x \in \mathcal{X}} \left| \hat{P}_{x_1^n}(x) - p(x) \right| \leq \delta \right\}.$$

Here,

$$\hat{P}_{x_1^n}(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{x_i = x\}$$

is the empirical distribution (type) of the sequence x_1^n .

Strong Typicality (Alternative Definition)

AEP

Weak Typicality

AEP Properties

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Joint Typicality

Strong typicality is sometimes defined coordinate-wise as

$$\left| \hat{P}_{X_1^n}(x) - p(x) \right| < \epsilon \quad \text{for every } x \in \mathcal{X},$$

which is an ℓ_∞ constraint on the empirical distribution.

Equivalence (Finite Alphabet). For $|\mathcal{X}| = m < \infty$, the two definitions are equivalent up to rescaling of parameters since

$$\|\hat{P}_{X_1^n} - p\|_\infty \leq \|\hat{P}_{X_1^n} - p\|_1 \leq m \|\hat{P}_{X_1^n} - p\|_\infty.$$

Thus, both definitions describe empirical distributions close to $p(x)$, and generate asymptotically equivalent typical sets.

Properties of Strong AEP

AEP

Weak Typicality

AEP Properties

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Joint Typicality

Theorem (Strong AEP)

There exists $\eta > 0$ such that $\eta \rightarrow 0$ as $\delta \rightarrow 0$, and the following hold

1. *If $x_1^n \in T_\delta^{(n)}$, then $H(X) - \eta \leq -\frac{1}{n} \log p(x_1^n) \leq H(X) + \eta$.*
2. *$\Pr \left\{ T_\delta^{(n)} \right\} > 1 - \delta$ for n sufficiently large.*
3. *$\left| T_\delta^{(n)} \right| \leq 2^{n(H(X) + \eta)}$.*
4. *$\left| T_\delta^{(n)} \right| \geq (1 - \delta) 2^{n(H(X) - \eta)}$ for n sufficiently large.*

Strong Typicality vs Weak Typicality

AEP

Weak Typicality

AEP Properties

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Joint Typicality

- Weak typicality (entropy typicality): empirical entropy $\approx$ true entropy.
- Strong typicality (letter typicality): empirical distribution $\approx$ true distribution.
- Strong typicality $\implies$ Weak typicality, but not vice versa.
- Strong typicality works only for finite alphabet, i.e., $|\mathcal{X}| < \infty$.

Empirical Entropy, Self-Information, and KL Divergence (1/2)

AEP

Weak Typicality

AEP Properties

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Joint Typicality

Normalized self-information:

$$-\frac{1}{n} \log p(X_1^n) = -\frac{1}{n} \sum_{i=1}^n \log p(X_i) = -\sum_{x \in \mathcal{X}} \hat{P}_{X_1^n}(x) \log p(x)$$

Sample average = average under the empirical distribution:

$$\boxed{\frac{1}{n} \sum_{i=1}^n f(X_i) = \sum_{x \in \mathcal{X}} \hat{P}_{X_1^n}(x) f(x),} \quad \text{with } f(x) = \log p(x).$$

Proof:

$$\sum_{x \in \mathcal{X}} \hat{P}_{X_1^n}(x) f(x) = \sum_{x \in \mathcal{X}} \left(\frac{1}{n} \sum_{i=1}^n \mathbf{1}\{X_i = x\} \right) f(x) = \frac{1}{n} \sum_{i=1}^n f(X_i).$$

Empirical Entropy, Self-Information, and KL Divergence (2/2)

AEP

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Joint Typicality

Cross-entropy decomposition:

$$-\frac{1}{n} \log p(X_1^n) = - \sum_x \hat{P}_{X_1^n}(x) \log p(x) = H(\hat{P}_{X_1^n}) + D_{\text{KL}}(\hat{P}_{X_1^n} \| p)$$

This identity is the **bridge between probability, combinatorics, and information**.

Interpretation:

- $H(\hat{P}_{X_1^n})$: empirical (sample) entropy
- $D_{\text{KL}}(\hat{P}_{X_1^n} \| p)$: divergence from the true distribution

Why This Identity Is Fundamental

AEP

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1. AEP (entropy concentration)

- By the WLLN, $\hat{P}_{X_1^n} \rightarrow p \implies D_{\text{KL}}(\hat{P}_{X_1^n} \| p) \rightarrow 0$ and $H(\hat{P}_{X_1^n}) \rightarrow H(p)$.
- Therefore, $-\frac{1}{n} \log p(X_1^n) \rightarrow H(p)$.

2. Method of types (exact counting)

- All sequences with the same empirical distribution (type) have the same probability.
- The number of sequences of type q satisfies

$$|T(q)| \approx 2^{nH(q)}.$$

- Entropy emerges as the exponent governing combinatorial growth.

Sanov's Theorem

AEP

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Sanov's theorem (large deviations)

- The probability that $\hat{P}_{X_1^n}$ is close to $q \neq p$ decays exponentially:

$$\Pr\{\hat{P}_{X_1^n} \approx q\} \approx 2^{-nD_{\text{KL}}(q\|p)}.$$

- KL divergence governs the decay rate of atypical events.

Unifying view: Entropy controls *how many* sequences exist; KL divergence controls *how unlikely* they are.

High-probability Set (1/2)

AEP

Weak Typicality

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Joint Typicality

To this end, we know that the $A_\epsilon^{(n)}$ is a fairly small set that has most of the probability.

Question: Is it the smallest set with such a property?

Definition

For $\delta > 0$, let $B_\delta^{(n)} \subset \mathcal{X}^n$ be the smallest set such that $\Pr(X_1^n \in B_\delta^{(n)}) \geq 1 - \delta$.

Theorem

Let $\delta < \frac{1}{2}$. For any $\delta' > 0$,

$$\frac{1}{n} \log |B_\delta^{(n)}| > H - \delta'$$

for n sufficiently large.

High-probability Set (2/2)

AEP

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Typical set vs High-probability set.

- For sufficiently large n (depending on δ and δ'), $B_\delta^{(n)}$ has at least $2^{n(H-\delta')}$ elements. The ϵ -typical set $A_\epsilon^{(n)}$ has about $2^{n(H\pm\epsilon)}$ elements.
- $A_\epsilon^{(n)}$ and $B_\delta^{(n)}$ have roughly the same number of elements to first order in the exponent.
- $B_\delta^{(n)}$ is defined via minimal cardinality, not typicality.

Encoding for the Typical Set (1/2)

AEP

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The fact that the typical set has probability approaching 1 as n grows large means that we “only need” to care about encoding sequences in the typical set.

The number of bits required to encode a set $\mathcal{S}$ is $\lceil \log |\mathcal{S}| \rceil$, where $\lceil a \rceil$ denotes the smallest integer greater than or equal to a .

Let $\{X_i\}_{i=1}^n$ be i.i.d. with $X_i \sim p(x)$. Consider the following scheme for coding $x_1^n \in \mathcal{X}^n$.

- First, impose a total order on all sequences in $A_\epsilon^{(n)}$ and its complement according to some criterion (e.g., lexicographic order: “ABC, ACB, BAC, BCA, CAB, CBA”).
- Use the first bit as an indicator of typicality: start with 0 if $x_1^n \in A_\epsilon^{(n)}$, and 1 otherwise.

Encoding for the Typical Set (2/2)

- If $x_1^n \in A_\epsilon^{(n)}$, since $|A_\epsilon^{(n)}| \leq 2^{n(H(X)+\epsilon)}$, use $n(H(X) + \epsilon) + 1$ bits for encoding (the additional 1 bit is due to integrality),
- If $x_1^n \notin A_\epsilon^{(n)}$, use no more than $n \log |\mathcal{X}| + 1$ bits to encode it.

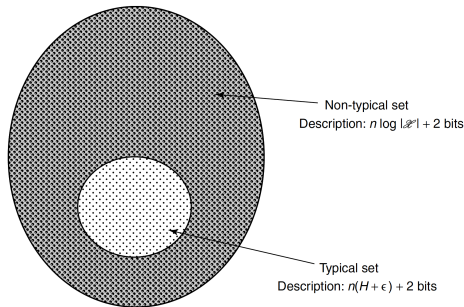


Figure: Encoding for the typical set.

Consequences of AEP (1/2)

Let $\ell(x_1^n)$ denote the length of the codeword (a binary string) corresponding to $x_1^n \in \mathcal{X}^n$. For a sufficiently large n , we have

$$\begin{aligned} \mathbb{E}[\ell(x_1^n)] &\leq P\left(x_1^n \in A_\epsilon^{(n)}\right) \times (n(H + \epsilon) + 2) + P\left(x_1^n \notin A_\epsilon^{(n)}\right) \times (n \log |\mathcal{X}| + 2) \\ &= 2 + P\left(x_1^n \in A_\epsilon^{(n)}\right) \times (n(H + \epsilon)) + P\left(x_1^n \notin A_\epsilon^{(n)}\right) \times (n \log |\mathcal{X}|) \\ &\leq 2 + n(H + \epsilon) + \epsilon n \log |\mathcal{X}| =: n(H + \tilde{\epsilon}) \end{aligned}$$

where $\tilde{\epsilon} = \epsilon(1 + \log |\mathcal{X}|) + \frac{2}{n}$ can be arbitrarily small by appropriate choices of ϵ and n .

Remark:

1. This is a proof of existence, not a practical coding scheme.
2. The code is not prefix-free.
3. This motivates Huffman, arithmetic coding, etc.

Consequences of AEP (2/2)

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Joint Typicality

Theorem ($H(X)$ bits are needed to encode X_1^n per symbol on average)

Consider i.i.d. $\{X_i\}_{i=1}^n \sim p(x)$. Let $\epsilon > 0$, then there exists a code that maps sequences x_1^n into binary strings, such that the mapping is one-to-one and $E \left[\frac{1}{n} \ell(x_1^n) \right] \leq H(x) + \epsilon$ for n sufficiently large.

The above theorem explains the achievability part of the *Source Coding Theorem*:

- A sequence of symbols can be compressed to a binary string with an average of $H(X)$ bits per symbol.
- This further reinforces the interpretation of the entropy as the average information content of a random source.

Jointly Typical Sequences

AEP

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Joint Typicality

Two sequences x_1^n and y_1^n are jointly ϵ -typical if

1. the pair (x_1^n, y_1^n) is ϵ -typical with respect to the joint distribution $p(x_1^n, y_1^n) = \prod_{i=1}^n p(x_i, y_i)$ (i.e., pairwise independence).
2. both x_1^n and y_1^n are ϵ -typical w.r.t. their marginal distributions $p(x_1^n)$ and $p(y_1^n)$.

Remark: For i.i.d. sources, marginal typicality follows from joint typicality.

Jointly Typical Set

AEP

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The set of all such pairs of sequences (x_1^n, y_1^n) is denoted by

$$A_{\epsilon}^{(n)}(X, Y) = \left\{ (x_1^n, y_1^n) \in \mathcal{X}^n \times \mathcal{Y}^n : \begin{aligned} &\left| -\frac{1}{n} \log p(X_1^n) - H(X) \right| < \epsilon, \\ &\left| -\frac{1}{n} \log p(Y_1^n) - H(Y) \right| < \epsilon, \\ &\left| -\frac{1}{n} \log p(X_1^n, Y_1^n) - H(X, Y) \right| < \epsilon \end{aligned} \right\}$$

Joint AEP (1/2)

AEP

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Theorem (see the proof on page 196–198 of Cover's book)

Let (X_1^n, Y_1^n) be sequences of length n drawn i.i.d. $\sim p(x_1^n, y_1^n) = \prod_{i=1}^n p(x_i, y_i)$. Then,

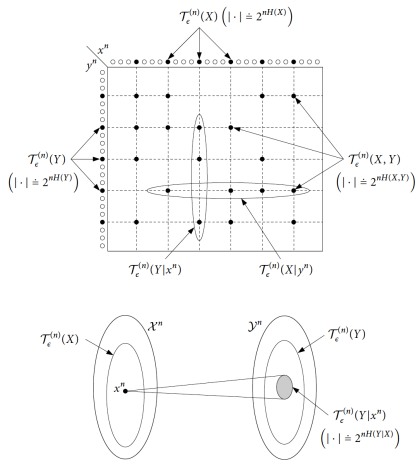
1. $\Pr \left((X_1^n, Y_1^n) \in A_\epsilon^{(n)} \right) \rightarrow 1$ as $n \rightarrow \infty$.
2. $|A_\epsilon^{(n)}| \leq 2^{n(H(X,Y)+\epsilon)}$.
3. If $(\tilde{X}_1^n, \tilde{Y}_1^n) \sim p(x_1^n)p(y_1^n)$, then

$$(1 - \epsilon)2^{-n(I(X;Y)+3\epsilon)} \leq \Pr \left((\tilde{X}_1^n, \tilde{Y}_1^n) \in A_\epsilon^{(n)} \right) \leq 2^{-n(I(X;Y)-3\epsilon)},$$

where the upper bound holds for n sufficiently large.

We use the joint AEP and random coding to prove the channel coding theorem (Chap 7).

Joint AEP (2/2)



Implication:

- Typical sets $|\mathcal{X}_1^n| \approx 2^{nH(X)}$ and $|\mathcal{Y}_1^n| \approx 2^{nH(Y)}$.
- Not all pair of typical X_1^n and typical Y_1^n are jointly typical: only about $2^{nH(X, Y)}$.
- Intuitive argument for joint typicality lemma: the probability of any randomly chosen pair is jointly typical is about

$$\frac{2^{nH(X, Y)}}{2^{n(H(X) + H(Y))}} = 2^{-nI(X; Y)}.$$

Figure: Gamal–Kim Book